

Engida Cloud HMS

Architecture, tenancy, and every state machine from the spec — drawn. እንግዳ = guest.

Stack React · Supabase · Postgres 15 **Market** Ethiopia · ETB · GC/EC

Isolation tenant_id + RLS **Dialogs** D.001 ... D.082

SHARED STATUS SYSTEM — READ ONCE, APPLIES TO EVERY MACHINE BELOW

Start / neutral Live · active · clean In progress · hold
Attention · on-hold · past-due Terminal · completed Terminal · adverse

6 System architecture

One Supabase project per environment. The database is the only source of truth — RLS on every table, RPCs as state machines, triggers writing the audit log. Edge Functions exist only to talk to the outside world.

§2 CLIENTS → SUPABASE (PER ENV) → EXTERNAL SERVICES

CLIENTS

Staff Web App · React SPA Super-Admin Console · /platform Guest Portal · P2

HTTPS · supabase.js v2

SUPABASE PROJECT — ONE PER ENVIRONMENT

Auth · GoTrue

email + OTP + TOTP
custom claims hook
{tenant_id, roles}

Postgres 15

- app schema (55 tables)
- RLS enabled + forced
- RPC fns = state machines
- pg_cron scheduled jobs
- triggers → audit_log

Storage

buckets (path RLS)
ids · invoices · wo

Realtime

board channels
cache.invalidate

Edge Functions · Deno — external I/O only

chapa.webhook telebirr.webhook notify.dispatch invoice.pdf ical.sync

tenant.export backup.export saas.dunning · cron

outbound · allow_listed_hosts_only

Platform services

Resend (email) · Sentry (errors)

Payment & channel rails

Chapa · Telebirr · SMS gateway · OTA
iCal endpoints



Multi-tenancy hierarchy

Shared-schema isolation. Every operational row carries **tenant_id** and **property_id**, sourced from the JWT — never from the client — so cross-tenant access is impossible by construction.

§4 PLATFORM → TENANT → PROPERTY → DATA · RLS AT EVERY LEVEL

Platform

plan catalog · feature flags · audit · backups

JWT.scoped

TENANT = HOTEL COMPANY

Tenant

subscription · plan · status (trial → ... → purged)

tenant_id

PROPERTY = ONE HOTEL

Property

business_date · tax config · invoice sequence · cashier shifts

+ property_id

Memberships — user × tenant × property[] × roles[]

Inventory — rooms · room_types · rate_plans ·
rates

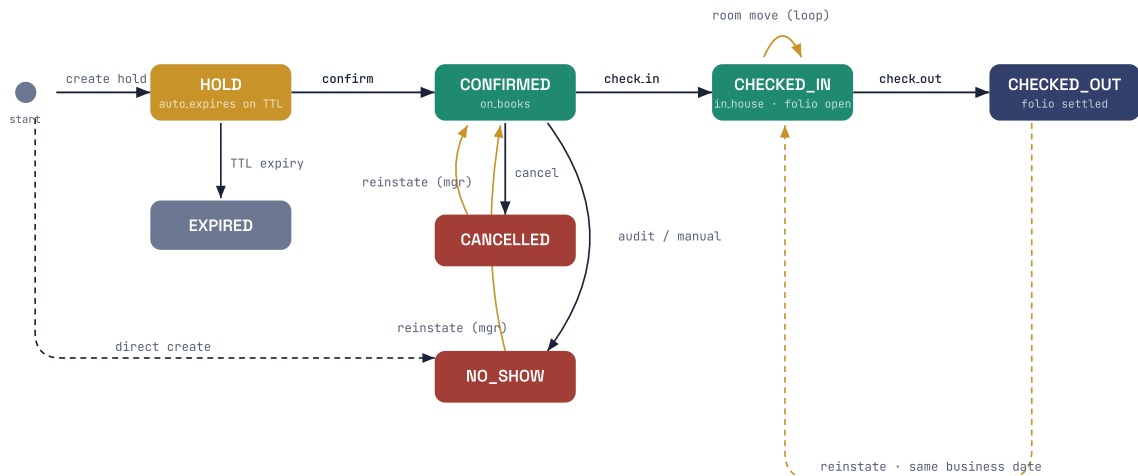
Operational data — every row stamped tenant_id +
property_id



Reservation lifecycle

The core booking machine (§13). Holds expire on a TTL; managers can reinstate from adverse terminals; a same-day reinstate can even reopen a checked-out folio. Every transition runs server-side through an RPC.

§13.1 STATES · TRANSITIONS · ACTORS



- system / staff transition - - - direct create · same-day
- manager-only (reinstatement / room move)



Availability formula

Computed live per room-type per date (§13.3). Holds count against inventory so two guests can't race the same room; the overbooking allowance is the only way to go below zero, and it's manager-gated.

§13.3 AVAILABLE(TYPE, DATE)

available(type, date) =

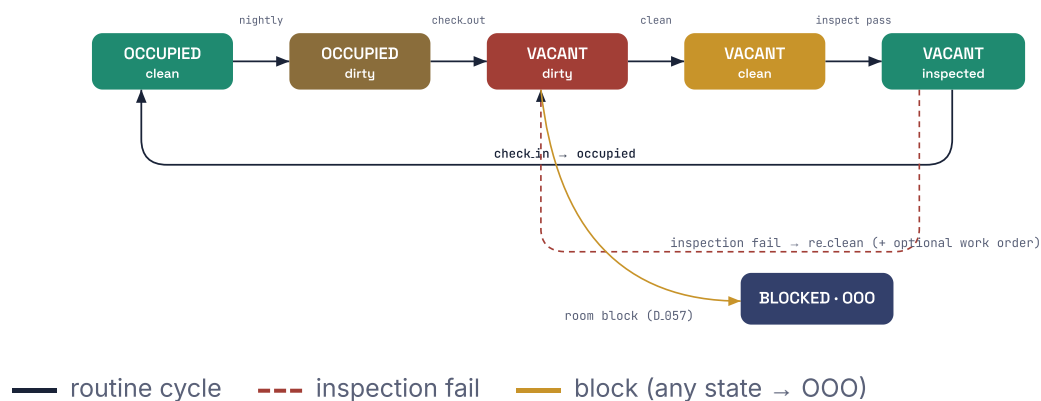
- +** `physical_rooms(type)` active, not soft.deleted
- `000_blocked(type, date)` room_blocks · kind = '000'
- `sold(type, date)` reservation_rooms in {hold, confirmed, checked_in} covering the date — holds included
- +** `overbook_allowance(type, date)`



Housekeeping status machine

Per-room status cycle (§16.1). Check-out dirties a room; a housekeeper cleans, a supervisor inspects; check-in occupies it again. An OOO block can pull any room out of the cycle and suppresses its tasks.

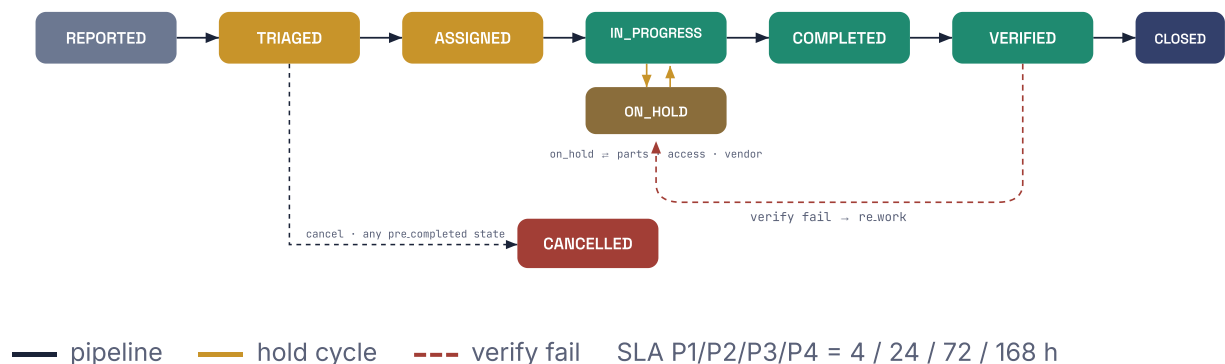
§16.1 ROOM HK_STATUS · CYCLICAL



Work-order lifecycle

Maintenance pipeline (§17.1). Priority set at triage drives the SLA clock (P1-P4 = 4 / 24 / 72 / 168 h). Verification can bounce a job back; a breach escalates to the manager via the `sla_watch` cron.

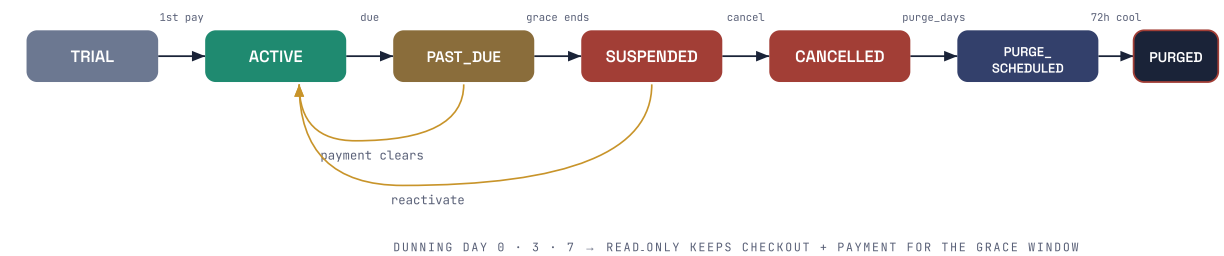
§17.1 REPORTED → ... → CLOSED · SLA-DRIVEN



SaaS tenant lifecycle

Billing state for the whole tenant (§28). Dunning fires on day 0/3/7; a past-due or suspended tenant enters a humane read-only mode that still lets staff check guests out and take payment for a grace window before anything is locked.

§28 TRIAL → ... → PURGED · DUNNING & READ-ONLY



— lifecycle progression — recovery (pay / reactivate)

read-only = humane grace, not a hard lock



Application navigation map

The staff app shell (§35). Modules marked ★ are plan-gated; /platform is a separate console visible only to platform roles.

§35 MODULE MAP · ROUTE GUARDS BY PERMISSION

Dashboard KPI strip arrival/departure queues	Front Desk Arrivals · Departures In.House Tape Chart	Reservations List · Calendar Groups ★
Guests CRM · dedupe DNR · privacy	Housekeeping Board · Tasks Lost & Found	Maintenance Work orders 000 / 00S blocks
POS ★ Outlets Day Close	Billing Folios · Invoices Payments · AR · Shifts	Rates Calendar · Plans Restrictions
Reports flash · forecast	Channels ★ iCal setup	Settings property · tenant

CSV / PDF export

conflict queue

team · flags

Night Audit ►

6.step wizard

rolls business_date

/platform

Tenants · Plans

Billing · Health

Backups · Announce

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Engida Cloud HMS · visual companion to the master blueprint v1

Diagrams cover §2 · §4 · §13 · §16 · §17 · §28 · §35 – every ASCII figure and prose state machine, drawn.